

PostgreSQL Research Outline for GMAP (Boeing 1)

Summary: PostgreSQL is a robust open-source relational database management system. Its strengths include reliability and data integrity, concurrency control using multi-version concurrency control, and strong compliance standards.

Notable Functionalities:

- Data storage and retrieval
- Concurrency and transaction management
- Indexing
- Data integrity and security
- Advanced types (JSON/JSONB, UUID, enumerated types, etc)
- Built-in language along with Python
- High availability
- Interactive terminal (psql) along with management tools (pgAdmin)
- Cloud capabilities (Azure, AWS, Google Cloud SQL)
- Supports web and enterprise applications (Node.js, Rails, etc)

How we plan to use it:

In addition to implementing our database design structure we plan to leverage it's multi-tenant capabilities to enhance our data isolation and scalability. Furthermore, we can use their various data integrity and security functionalities along with leveraging their ability to use advanced types such as JSONB. We will use both the CLI tools and management tools to meet our project database requirements.